Beyond Fake or Real: Evidence-Grounded, Explainable Five-Class News Verification with Temporal Context Checking

Navishka Sharma, Naitik Kukreja, Naveen, Prateek Srivastava, and Nikhil
Department of Computer Science and Engineering (AI/ML)
KIET Group of Institutions, Ghaziabad, Uttar Pradesh, India
{navishka.sharma, naitik.kukreja, naveen, prateek.srivastava, nikhil}@kiet.edu

Abstract—Most automatic "fake news" detectors are binary text classifiers: they learn how fabricated articles tend to be written rather than whether the claims in them are true, so they cannot explain a verdict, cannot describe an item that is only partly wrong or true-but-out-of-date, and cannot admit that they do not know. We present a compact, fully open-source verification pipeline that can run on a student laptop. Given a headline, an article or a URL, the system extracts check-worthy claims, retrieves evidence from free online sources and from an offline index of fact-checked statements, labels the stance of every evidence passage with a natural-language-inference cross-encoder, compares publication dates to detect old news recirculated as "breaking", and fuses these signals with transparent rules into one of five verdicts—Real, Fake, Partially True, Misleading or Unverifiable—together with a confidence score, dated evidence cards and a plain-language explanation. A fine-tuned DistilBERT content classifier with LIME token attributions contributes only to the confidence estimate. A cross-dataset audit shows why the in-domain accuracies near 99 % reported on standard corpora do not transfer. Code, data splits and the claim set are released.

Index Terms—fake news detection, fact verification, evidence retrieval, stance detection, natural language inference, explainable AI, temporal misinformation

I. Introduction

News now reaches most people through feeds and messaging groups rather than through editors, and the damage done by what slips through is well documented. Tracing roughly 126,000 story cascades on Twitter over eleven years, Vosoughi et al. found that false stories were about 70 % more likely to be retweeted than true ones and that the truth needed roughly six times longer to reach 1,500 people [1]. The World Economic Forum's Global Risks Report 2024 places misinformation and disinformation first among global risks for the coming two years, ahead of extreme weather and armed conflict [2]. A study of the COVID-19 "infodemic" linked a single rumour—that drinking concentrated alcohol or methanol cures the virus—to about 800 deaths and roughly 5,800 hospital admissions in the first three months of 2020 [3]. India, the setting of this work, offers its own examples: child-kidnapping rumours forwarded on WhatsApp were linked to at least 17 mob killings across the country between April and July 2018 [4], [5], and Indian fact-checkers routinely find genuine footage of earlier floods, riots and accidents re-captioned as "breaking" news for a new event—in one analysis of 125 fact-checked Indian COVID-19 items, nearly half paired text with video and 94 % originated on social platforms [6]. This last pattern—true material presented in the wrong time or place—is one that most detectors are not built to see.

Manual fact-checking is accurate but slow. Most academic work on automatic detection has therefore taken a shortcut: treat the task as binary text classification and let a model learn the surface cues of fabricated writing. Such classifiers report accuracies above 95 % on benchmark corpora [7], [8], but the numbers hide three weaknesses. First, they learn style and dataset artefacts rather than facts: a corpus whose genuine articles all come from one wire service can be separated almost perfectly by boilerplate alone [9], and the same models lose much of their accuracy on a different corpus. Second, a binary label cannot express the shapes real misinformation takes—a true event with a wrong number, a real quotation attached to the wrong occasion, a 2018 video captioned as today's news. Third, a classifier always produces an answer; it cannot say that the evidence is insufficient, which for hyper-local or very recent claims is the only honest response.

Problem statement. Given a news item—a headline, an article text or a URL—automatically determine, with supporting evidence, dates and a human-readable explanation, whether its central claims are real, fake, partially true, misleading or unverifiable, using only free tools and hardware available to students.

We address this problem by moving the decision from the classifier to the evidence: the system selects the check-worthy sentences of an item, retrieves passages from free web search, Wikipedia, the Google Fact Check Tools API and an offline index of fact-checked statements, labels each passage as supporting, refuting or neutral, compares the dates involved, and applies a small, inspectable set of rules. A style-based classifier is retained, but it may only adjust the confidence or break a single-source tie; it never outputs Fake or Real on its own when evidence exists. The contributions of this paper are:

- An evidence-grounded five-class verdict scheme with

abstention. We define Real, Fake, Partially True, Misleading and Unverifiable operationally, map LIAR's six PolitiFact grades onto them, and reach each verdict through a published decision table whose confidence reflects evidence agreement, coverage and source reliability, so every output carries the rule that fired, the evidence used and an explicit uncertainty statement (objective O6).

- A laptop-scale claim-to-verdict pipeline. Claim identification, evidence retrieval from free sources with an offline FAISS fallback, NLI stance detection with a small cross-encoder, and a temporal/context check that flags old news presented as new and entity–date mismatches, all running on CPU within seconds offline (objectives O3–O5).
- An explainable content classifier evaluated honestly. A fine-tuned DistilBERT model with LIME token attributions serves as the style signal; we report in-domain results on WELFake, a cross-dataset transfer and leakage audit against ISOT, and an ablation of how much the classifier adds to the evidence-based verdict (objective O1).

The rest of the paper is organised as follows. Section II surveys related work and states the gap; Sections III–V describe the datasets, the pipeline with its fusion rules, and the experimental setup; Section VI reports results, Section VII compares them with existing systems, and Section VIII concludes.

II. Related Work

We group prior work into five themes that correspond to the components of our system; Table I summarises the most relevant papers along the dimensions that matter to us.

A. Datasets and Benchmarks

LIAR [10] collects 12.8k short political statements rated by PolitiFact on a six-point scale from pants-on-fire to true. Its best text-and-metadata model reached only about 27 % six-way accuracy, which remains a useful reminder that fine-grained truthfulness cannot be read off the wording of a sentence. ISOT [9], [11] pairs 21k Reuters articles with 23k articles from sites flagged by fact-checkers; because every genuine article shares one source, the corpus is easy to separate and is widely suspected of source leakage. WELFake [7] merges four public corpora into 72k articles to reduce exactly this kind of overfitting and is our primary training set; one of its four sources is ISOT itself, and our preprocessing audit found that almost every ISOT article recurs verbatim in WELFake (Section III), which is why our cross-corpus experiment removes the shared articles from the WELFake side. FakeNewsNet [12] adds social context and PolitiFact/GossipCop labels to news items; we use its PolitiFact titles in our offline index. On the verification side, FEVER [13] contributes 185k claims written against Wikipedia and labelled Supports, Refutes or Not Enough Info, defining the retrieve-then-verify task that our stance module inherits, while MultiFC [14] gathers 35k claims with the heterogeneous verdict labels of 26 real fact-checking sites. ClaimBuster [15] provides 23.5k debate sentences annotated for check-worthiness. Surveys by Zhou and Zafarani [16] and Guo et al. [17] organise this landscape: the former distinguishes knowledge-, style-, propagation- and source-based detection, and the latter formalises fact-checking as claim detection, evidence retrieval, verdict prediction and justification production—the sequence our pipeline follows.

B. Content-Based Classifiers

The earliest and still most common approach treats detection as document classification. Ahmed et al. [9] reported that TF-IDF n-grams with a linear SVM reach roughly 92 % accuracy on ISOT, and Pérez-Rosas et al. [18] found linear models on lexical, syntactic and readability features to be competitive on crowd-sourced fake news. Pre-trained transformers [19] raised these numbers further: FakeBERT [8] feeds BERT embeddings through parallel convolutional blocks and reports 98.9 % accuracy on a Kaggle corpus, and WELFake's own linguistic-feature model reports 96.7 %. DistilBERT [20] keeps about 97 % of BERT's language-understanding performance at 40 % fewer parameters, which is what makes fine-tuning on a 4 GB consumer GPU feasible and is why we adopt it. A recent benchmark on raw WELFake reports 0.954 macro-F1 for fine-tuned DistilBERT and 0.968 for RoBERTa [21]. Blackledge and Atapour-Abarghouei [22] are a rare exception to in-domain-only evaluation: DistilBERT, BERT and DeBERTa reach 0.989–0.997 accuracy on an ISOT hold-out but only 0.670–0.697 when the ISOT-trained models are tested on another corpus. Apart from that study, what these papers share is an in-domain, binary evaluation without cross-corpus transfer, and none can tell a user why an article is suspicious beyond its wording. We keep such a classifier, but only as a secondary signal.

C. Claim Detection, Evidence Retrieval and Fact Verification

ClaimBuster [23] showed that check-worthy sentences in political debates can be ranked with sentence-level features and a supervised classifier, and its later benchmark release [15] is the standard training set for the task; our claim identifier combines similar hand-crafted cues with an optional classifier trained on that data. For verification, the FEVER baseline [13] chains document retrieval, sentence selection and an NLI model and scored 31.9 FEVER points; the shared-task winner, NSMN [24], unified the three steps with semantic matching networks, and KGAT [25] propagates evidence through a kernel-based graph attention network to exceed 70 FEVER points—the current ceiling for Wikipedia-grounded verification. DeClarE [26] verifies free-text claims against web search results with attention over the retrieved articles

and a source-credibility embedding, which is close in spirit to our online path. Two components make retrieval and stance practical at our scale: Sentence-BERT [27] bi-encoders (we use the 22M-parameter MiniLM variant [28]) turn semantic search over thousands of fact-checked statements into a millisecond nearest-neighbour lookup, while a small DeBERTaV3 cross-encoder [29] trained on SNLI and MultiNLI [30] provides an off-the-shelf three-way stance judgement on CPU. Hanselowski et al. [31] re-examined the Fake News Challenge and showed that headline-body stance systems that looked strong under the challenge metric had far lower macro-F1 on the minority agree/disagree classes—a caution we take into account by reporting macro-F1 throughout.

D. Explainability

LIME [32] explains an individual prediction by fitting a sparse linear surrogate on perturbed inputs, and SHAP [33] unifies several attribution methods under Shapley values; both are model-agnostic and produce token-level highlights that lay users can read. In the fake-news setting, dEFEND [34] co-attends over article sentences and user comments so that the sentences and comments that drove the decision can be shown as the explanation, and DeClarE [26] exposes the retrieved evidence words its attention focused on. These methods explain a model; they do not by themselves ground the verdict in checkable facts. We therefore treat LIME highlights as an explanation of the style signal only, and build the primary explanation from the retrieved evidence, its stance, its source and its date.

E. Temporal and Contextual Misinformation

Much misinformation is not fabricated text but genuine material displaced in time or place. Vosoughi et al. [1] quantified how much faster and deeper falsehoods travel, and Zubiaga et al. [35] survey rumour detection, tracking and stance classification as an inherently temporal process in which a story's veracity is resolved over time. In India, Al-Zaman [6] analysed 125 fact-checked COVID-19 stories and found that two-thirds concerned health, that text paired with video was the commonest format, and that almost all originated on social platforms rather than in mainstream media; reporting on the 2018 WhatsApp lynchings [4], [5] traced the killings to rumour messages and videos forwarded as local warnings. Existing detectors ignore publication dates entirely; we add a lightweight check that compares the input's date and recency cues against the dates of the supporting evidence and against the entities it mentions.

F. The Gap

Table I makes the gap visible. Style-based classifiers are accurate in-domain but produce no evidence, no partial-truth labels and no abstention; FEVER-style verifiers ground their decisions in evidence but assume Wikipedia-only, synthetic claims and do not consider time; explainability work explains models rather than facts. No system we are aware of combines (i) a five-way verdict with an explicit Unverifiable class, (ii) evidence retrieval from free sources with an offline fallback, (iii) a temporal and contextual check, and (iv) evidence-linked explanations, while running on ordinary student hardware. Those properties are the three contributions listed in Section I, and the rest of this paper builds and evaluates them.

TABLE I
Representative prior work compared on the properties this paper targets. Ev. = grounds the verdict in retrieved evidence; Exp. = explains the verdict to a user; Abst. = can abstain; Temp. = checks dates or context. Full table in docs/literature_table.md.

Work	Output	Ev.	Exp.	Abst.	Temp.
LIAR CNN [10]	6-way	–	–	–	–
ISOT SVM [9]	binary	–	–	–	–
WELFake [7]	binary	–	–	–	–
FakeBERT [8]	binary	–	–	–	–
FEVER baseline [13]	3-way	✓	–	NEI	–
KGAT [25]	3-way	✓	–	NEI	–
DeClarE [26]	binary	✓	✓	–	–
dEFEND [34]	binary	–	✓	–	–
MultiFC [14]	per-site	✓	–	–	–
This work	5-way	✓	✓	✓	✓

III. Datasets and Preprocessing

Table II lists the corpora and the role each plays. WELFake [7] (72,134 articles) is the primary training set for the content classifier and ISOT [9] (44,898 articles) the secondary one. LIAR [10] (12,836 PolitiFact statements, official splits) trains a three-way truthfulness head and, together with 1,056 FakeNewsNet [12] PolitiFact titles, populates the offline fact-check index. A balanced FEVER [13] subset (20,000 train + 3,000 dev claims) evaluates the stance model. LIAR's six grades map onto our scheme as true/mostly-true → Real, half-true → Partially True, barely-true → Misleading and false/pants-fire → Fake; the three-way collapse used for the LIAR head is TRUE / MIXED / FALSE. Unverifiable has no counterpart in any corpus—it arises only from retrieval failure at run time.

Live Claims Set. For end-to-end evaluation we built 95 recent claims (fact-checks published between February 2025 and 25 October 2026, 93 of them in 2026) from PolitiFact (46), Snopes (31), Lead Stories (9), BOOM (4), Alt News (3) and Full Fact (2). Each rating, claim text and date was read from the live fact-check page and mapped to the five classes (e.g. Snopes Miscaptioned/Outdated → Misleading); Unverifiable comes only from Lead Stories evidence-absence ratings. Class counts are Fake 27, Real 21, Partially True 19, Misleading 19, Unverifiable 9. A stratified 30-item dev slice (seed 42) is reserved for tuning fusion thresholds; the remaining 65 items are held out.

TABLE II
Datasets after preprocessing (make data).

Dataset	Rows	Labels	Role
WELFake	60,799	real / fake	classifier (primary)
ISOT	37,567	real / fake	leakage, cross-dom.
LIAR	12,836	$6 \to 3$-way	LIAR head, index
FEVER subset	23,000	S / R / NEI	stance evaluation
FakeNewsNet (PF)	1,056	real / fake	offline index
Live Claims (ours)	95	5-class	end-to-end (ESE)

Preprocessing. All corpora are converted to one schema (id, title, text, label, …, split). For WELFake and ISOT we (i) strip publisher and format boilerplate with 26 named text patterns and 2 title patterns—Reuters datelines, photo credits, captions, bylines, [VIDEO] tags, outlet suffixes, URL debris—while keeping content words such as "told Reuters"; (ii) drop empty and very short texts; (iii) de-duplicate greedily, first exact SHA-1 matches and then MinHash near-duplicates (Jaccard ≥ 0.9); and (iv) split 80/10/10, stratified, seed 42, giving 48,639 / 6,080 / 6,080 WELFake and 30,053 / 3,757 / 3,757 ISOT rows. Splits are stored as id lists and every model reads the same ids.

Leakage audit: ISOT $\subset$ WELFake. WELFake was assembled from four corpora, one of which is ISOT. Hashing normalised text shows that 37,542 of the 37,567 processed ISOT articles (99.93 %) occur verbatim in processed WELFake, so "train on WELFake, test on ISOT" is an in-domain experiment. For the cross-domain experiment we therefore use WELFake\ISOT: the WELFake rows not hash-matched to ISOT, minus 25 MinHash near-duplicates, i.e. 23,232 rows. A second audit trains a logistic regression on the 26 artefact-pattern counts alone (no words): it reaches 0.9973 five-fold accuracy on raw ISOT but 0.5568 on processed ISOT, which confirms that raw ISOT can be separated by source boilerplate alone and that our stripping removes that shortcut.

IV. Methodology

Fig. 1 shows the pipeline; every box is a module under src/ and runs behind one FastAPI endpoint, POST /analyze, in an online and an offline mode.

A. Claim Identification

The input is cleaned and sentence-split with spaCy en_core_web_sm. Sentences of fewer than six tokens, questions and imperatives are dropped; each remaining sentence s at position i receives the check-worthiness score

$$h(s) = 1.0\,[\mathrm{NE}] + 0.8\,[\mathrm{NUM}] + 0.6\,[\mathrm{REP}] + 0.5\,[i < 3] + 0.4\,[\mathrm{CMP}] - 0.8\,[\mathrm{OPN}] - 0.5\,[\mathrm{HDG}], \quad (1)$$

where the indicators denote a PERSON/ORG/GPE/EVENT/NORP/LOC entity, a CARDINAL/PERCENT/MONEY/DATE/QUANTITY entity, a reporting verb (said, announced, according to, …), a lead sentence, a number or comparative, a first-person opinion cue (I think, should, must) and a hedge (may, might, could). The score is normalised to $[0, 1]$ by its maximum (3.3) and clipped. Near-duplicate sentences (MiniLM cosine > 0.85) are removed and the top $k = 5$ are kept; a headline is always claim 1 with score 1. The optional ClaimBuster-trained classifier [15] is not used yet.

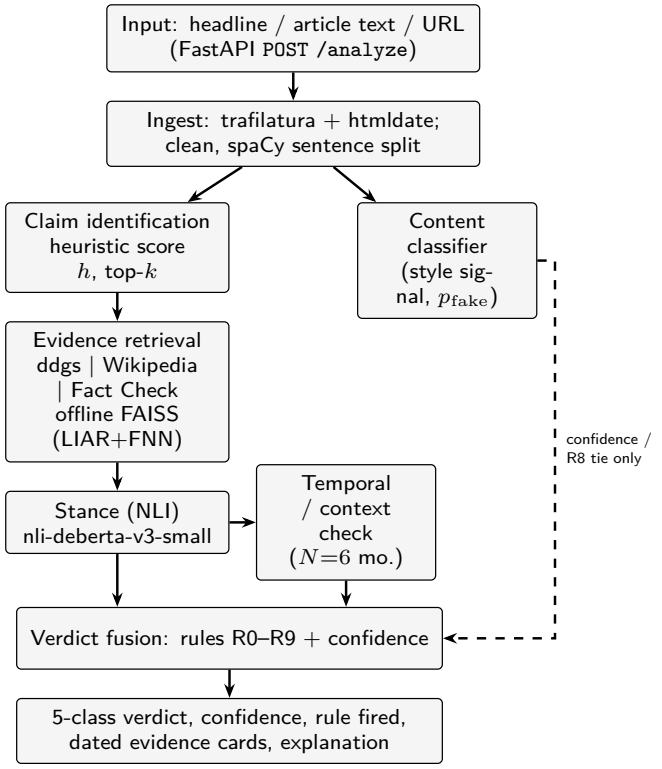


Fig. 1. System pipeline. The content classifier (dashed) influences only the confidence and the single-source tie-break R8; the verdict comes from evidence.

B. Evidence Retrieval

Online, each claim is sent to DuckDuckGo text search (ddgs), the Wikipedia search API (top page intros) and, when an API key is configured, the Google Fact Check Tools claims:search endpoint, each with a 6 s timeout. Result pages are fetched and extracted with trafilatura [36], dated with htmldate, and reduced to the three sentences most similar to the claim. All network answers are cached in SQLite so that runs are repeatable. Offline, the claim is embedded with all-MiniLM-L6-v2 [27], [28] and searched in a FAISS [37] IndexFlatIP over L2-normalised vectors (cosine) of all 12,836 LIAR statements and 1,056 FakeNewsNet titles (13,892 vectors, 20.35 MB); hits with cosine < 0.55 are discarded. Each item receives a source-reliability weight w_s from an editable domain list: fact-checkers 1.0, wire services and official sites 0.8, Wikipedia 0.6, unknown domains 0.3, satire/hoax sites 0 (flagged).

C. Stance Detection

Each (claim, evidence) pair is scored by the cross-encoder cross-encoder/nli-deberta-v3-small, a DeBERTaV3 [29] model trained on SNLI [38] and MultiNLI [30] only; we do not fine-tune it. With premise = evidence and hypothesis = claim, entailment, contradiction and neutral map to Supports, Refutes and Neutral with probability p. For items that carry a fact-checker rating (LIAR, FakeNewsNet, Google Fact Check) the rating is turned into a stance with $p = 0.95$, but only when the claim restates the rated statement (NLI entailment, or cosine ≥ 0.75 without contradiction). The signed contribution of an item is $\sigma(e) = \text{sign(stance)} \cdot p \cdot w_s$.

D. Temporal and Context Check

Three flags are computed for the main claim. Old news: the input contains a recency cue (breaking, just in, today, …) and the gap between the input date t_{post} and the (lower) median date of the supporting evidence is at least $N = 6$ months. Out of context: the claim's GPE/DATE/EVENT entities conflict with those of at least two supporting passages. Headline–body mismatch: NLI with the body as premise refutes the headline with $p \geq 0.7$.

E. Verdict Fusion and Confidence

For claim i let $S_i = \sum_e \sigma(e)$ and let E_i be the number of non-neutral items with $w_s \geq 0.5$. The main stance is Supports if $S_1 \geq \tau$, Refutes if $S_1 \leq -\tau$ and Neutral otherwise, with $\tau = 0.6$. Table III lists the decision rules, evaluated top to bottom (first match wins); the thresholds are those of src/fusion/fusion.yaml ($\tau = 0.6$, $N = 6$, $k = 5$, minimum evidence 2), which are defaults not yet tuned on the dev slice. The content classifier's fake probability p_{fake} appears in only two places: the R8 tie-break and a ± 0.10 confidence adjustment.

The confidence combines how one-sided, how broad and how reliable the evidence for the main claim is:

$$a = |S_1| \,/\, \left(\textstyle\sum_e p_e w_e + 10^{-6}\right), \quad c = \min(E_1, 4)/4, \tag{2}$$

$$q = \text{mean}(w_s \text{ over non-neutral items}), \tag{3}$$

$$\text{conf} = \text{clip}(0.5a + 0.3c + 0.2q + \delta,\ 0.05,\ 0.98), \tag{4}$$

where $\delta = +0.10$ if the classifier agrees with a Fake/Real verdict ($p_{\text{fake}} \geq 0.7$ / ≤ 0.3), -0.10 if it disagrees and 0 otherwise. For Unverifiable, conf $= 1 - \max(a, 0.2)$, i.e. how sure the system is that it cannot tell. R0 and the language check use fixed confidences of 0.9 and 0.8. Every response returns the rule that fired, the claims, the evidence cards with source, date, cosine and stance, and the temporal flags.

F. Content Classifiers

We train TF-IDF word n-gram models with multinomial Naive Bayes, logistic regression and a linear SVM [39], and fine-tune distilbert-base-uncased [20], [40] on title [SEP] text. A second DistilBERT head is fine-tuned on LIAR statements (text only, no speaker or party, which are label proxies) for TRUE / MIXED / FALSE with class-balanced cross-entropy. Until a full-data DistilBERT checkpoint exists, the deployed style signal is the TF-IDF + logistic-regression model.

TABLE III
Fusion rules (src/fusion/rules.py); first match wins. "Imp." = another claim with score $c_i \geq 0.6$.

Rule	Condition	Verdict
LANG	input is not English	Unv.
R0	input domain on the satire / hoax list	Fake
R1	no evidence, all evidence neutral, or $E_1 = 0$	Unv.
R2	Tier-1 fact-check rated false, stance Refutes, cosine to main claim ≥ 0.75	Fake
R3	main Refutes and $E_1 \geq 2$	Fake
R4	main Supports and a temporal/context flag	Misl.
R5	main Supports, an imp. claim has $S_i \leq -\tau$, $E_i \geq 2$	Part.
R6	main Neutral, imp. claims split (one $S_i \geq \tau$, one $S_i \leq -\tau$, each $E_i \geq 2$)	Part.
R7	main Supports and $E_1 \geq 2$	Real
R8	main non-neutral and $E_1 = 1$: Fake if Refutes and $p_{\text{fake}} \geq 0.85$; Real if Supports and $p_{\text{fake}} \leq 0.15$; else	Unv.
R9	anything else	Unv.

V. Experimental Setup

Splits and seed. WELFake and ISOT use the stratified 80/10/10 split (seed 42); LIAR uses its official 10,269 / 1,284 / 1,283 split; the FEVER subset uses 3,000 balanced dev claims. Model selection always uses validation macro-F1; the test split is used once per model.

Classical models. GridSearchCV with stratified 5-fold CV on the WELFake training split (48,639 rows), scoring macro-F1, over n-gram range $\{(1,1),(1,2)\}$ × max. features $\{50\text{k}, 100\text{k}, 200\text{k}\}$ × sublinear TF $\{\text{T}, \text{F}\}$ × $C \in \{0.1, 1, 10\}$ (LR, SVM; 36 configurations each) or $\alpha \in \{0.1, 0.5, 1.0\}$ (NB; 18). Winners (best CV macro-F1): NB $\alpha = 0.1$, 200k features (0.8684); LR $C = 10$, 100k features (0.9428); SVM $C = 1$, 200k features (0.9459); all with $(1,2)$-grams and sublinear TF. The winners are refitted on the full training split of each dataset.

DistilBERT. Planned grid: learning rate $\{2, 3, 5\} \times 10^{-5}$ × max. length $\{256, 512\}$ on a 20,000-row stratified subset of WELFake train, 3 epochs with the best epoch kept, AdamW with linear schedule, warmup ratio 0.1, weight decay 0.01, fp16, batch 16 at length 256 (8 × 2 gradient accumulation at 512), evaluated on the full validation split (6,080). The three length-256 runs were completed; the length-512 runs and the full-data retrain of the winner were stopped on a team decision and are deferred to the final evaluation. LIAR head: learning rate $\{2, 3\} \times 10^{-5}$ × epochs $\{3, 5\}$, max. length 64, batch 16, same optimiser settings.

TABLE IV
Tuned TF-IDF classifiers, test split (in-domain). Val = validation macro-F1.

Data	Model	Val	Acc	P	R	F1	AUC
WELFake	NB	0.8728	0.8771	0.8744	0.8782	0.8758	0.9443
	LR	0.9455	0.9462	0.9457	0.9449	0.9453	0.9875
	SVM	0.9495	0.9490	0.9485	0.9478	0.9481	0.9884
ISOT	NB	0.9550	0.9470	0.9458	0.9472	0.9464	0.9865
	LR	0.9892	0.9848	0.9854	0.9839	0.9846	0.9986
	SVM	0.9900	0.9878	0.9882	0.9870	0.9876	0.9990

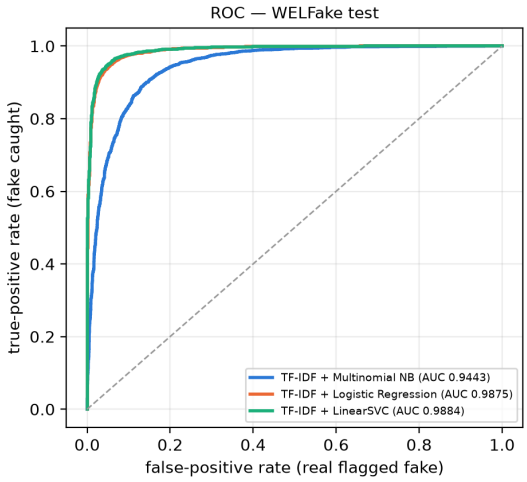


Fig. 2. ROC curves of the tuned classifiers on the WELFake test split.

Hardware and software. One laptop with an NVIDIA GeForce RTX 2050 (4 GB), 8 CPU threads and 13 GB RAM; the stance evaluation and the API run on CPU. Python 3.11, PyTorch 2.14.1, transformers 4.57.6, scikit-learn 1.9.1, sentence-transformers 5.7.0, faiss-cpu 1.15.1, spaCy 3.8.16, FastAPI 0.162.0. Metrics: accuracy, macro precision/recall/F1 and ROC-AUC (SVM from decision margins).

VI. Results

A. Content Classifiers (In-Domain)

Table IV gives held-out test results of the tuned classical models. On WELFake the linear SVM is best (macro-F1 0.9481, ROC-AUC 0.9884, Fig. 2), logistic regression is within 0.003, and Naive Bayes trails by about 0.07. Validation–test gaps are at most 0.0086, so the grid search did not over-fit the validation split. ISOT scores (up to 0.9876) are higher, but ISOT is in-domain here and is single-source on the real side. No DistilBERT test number exists yet (Section VI-C); objective O1 (DistilBERT macro-F1 ≥ 0.95 on WELFake test) is therefore not yet checked.

TABLE V
Cross-domain macro-F1, test splits (rows = train, columns = test). Proc. = artefacts stripped; raw = artefacts kept.

		Processed		Raw	
Model	Train	W\I	ISOT	W\I	ISOT
LR	W\I	0.9583	0.6638	0.9645	0.7616
	ISOT	0.5502	0.9846	0.5663	0.9933
SVM	W\I	0.9601	0.6611	0.9676	0.7611
	ISOT	0.5480	0.9876	0.5601	0.9949

TABLE VI
DistilBERT sweep on WELFake (20k-row training subset, full validation split, 3 epochs, best epoch 3 in all runs).

lr	max_len	val macro-F1	val acc	min
2×10^{-5}	256	0.9869	0.9872	14.8
3×10^{-5}	256	0.9863	0.9865	14.3
5×10^{-5}	256	0.9871	0.9873	15.2
$\{2, 3, 5\} \times 10^{-5}$	512	deferred to ESE		
full-train retrain	–	deferred to ESE		

B. Cross-Domain Transfer and Artefacts

Table V trains on one domain and tests on the other, using the disjoint WELFake\ISOT domain. Leaving the training domain costs 0.29–0.30 macro-F1 from WELFake\ISOT to ISOT and 0.43–0.44 in the other direction, so the near-perfect in-domain numbers say little about unseen sources. Evaluated on raw text with the artefacts kept, transfer looks better (e.g. 0.7616 instead of 0.6638 for LR, W\I $\rightarrow$ ISOT), i.e. part of what "transfers" is boilerplate. This is the empirical reason that, in our system, evidence and not the classifier decides the verdict.

Error analysis. The SVM misclassifies 310 of the 6,080 WELFake test articles (164 missed fakes, 146 false alarms). Assigned by fixed, ordered rules, 154 are style overlap (wire-style fakes or tabloid-style real articles), 71 opinion or first-person commentary, 68 cases where a source cue (e.g. "Reuters", "via") points the wrong way, 7 satire, 5 short texts, 4 republished wire copy on fake-labelled sites and 1 near-duplicate with a conflicting label—again errors of style, not of fact.

C. DistilBERT Sweep

Table VI lists every completed run. At length 256 the three learning rates lie within 0.0008 validation macro-F1 (0.9863–0.9871), all peaking at epoch 3; at 5×10^{-5} the training loss is lowest (0.0077) while validation loss rises from epoch 2 (0.0572) to 3 (0.0671), the onset of overfitting. These are validation numbers of models trained on a 20k subset; the length-512 runs, the full-data retrain and the test evaluation are deferred to the final evaluation, so no DistilBERT test result is reported.

D. LIAR Truthfulness Head

The best configuration (lr 3×10^{-5}, 3 epochs, best epoch 2, validation macro-F1 0.4773) reaches 0.4669 test

TABLE VII
LIAR official test split (1,283 statements).

Model	Labels	Acc	Macro-F1
Majority class	3	0.3749	–
DistilBERT 3-way head (ours)	3	0.4669	0.4667
Majority class	6	0.2081	–
DistilBERT 6-way twin (ours)	6	0.2689	0.2686
Wang (2017) text-only CNN	6	0.270	n/r
Wang (2017) hybrid CNN	6	0.274	n/r

TABLE VIII
Zero-shot stance on 3,000 balanced FEVER dev pairs.

Class	Precision	Recall	F1
Supports	0.8488	0.8870	0.8675
Refutes	0.6190	0.8710	0.7237
Neutral (NEI)	0.7701	0.4220	0.5452
Accuracy / macro-F1	0.7267 / 0.7121		

accuracy and 0.4667 macro-F1 on the three-way task, against 0.3749 for the majority class (Table VII). Its six-way twin with the same settings reaches 0.2689 accuracy, on par with Wang's text-only CNN (0.270) [10]: short statements carry little truth signal in their wording, which is why the head is only an auxiliary signal.

E. Stance Detection on FEVER

On 3,000 balanced FEVER dev pairs (gold evidence text from the FEVER 2017 Wikipedia dump; NEI claims paired with the sentences retrieved for them by the Papelo system [41]; premise prefixed with the page title) the cross-encoder reaches 0.7267 accuracy and 0.7121 macro-F1 (Table VIII, Fig. 3). This is a zero-shot result—the model was trained on SNLI and MultiNLI only, not on FEVER—and it misses our objective O4 (≥ 0.85). The shortfall is concentrated in NEI: on the gold Supports/Refutes pairs accuracy is 0.8790, but 479 of 1,000 NEI pairs are predicted Refutes, because the retrieved sentences concern the same entity and state a different fact. Without the page-title prefix accuracy falls to 0.6337. In the pipeline this error mode is damped by R3, which needs two reliable refuting sources; a Refutes threshold tuned on FEVER training pairs and an optional one-epoch fine-tune are planned.

F. End-to-End Evaluation

The offline pipeline runs end to end behind POST /analyze, and the 95-item Live Claims Set (Section III) is labelled and split. End-to-end five-class results, fusion-threshold tuning on the dev slice, the ablation (classifier only / +evidence / +temporal / offline only) and latency measurements are deferred to ESE.

G. Expected Outcomes for Deferred Experiments

Table IX states where the experiments deferred to ESE are expected to land. None of these values is measured; each is marked † and will be replaced by the ESE run. They are derived only from our own measurements in this section or from the project's target objectives, and no objective is claimed as met on their basis.

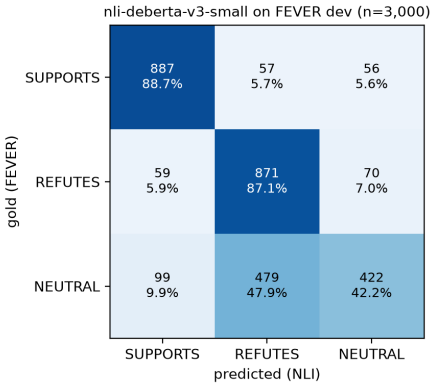


Fig. 3. Stance confusion matrix on the FEVER dev subset (rows = gold).

TABLE IX
Expected outcomes for experiments deferred to ESE (†expected, not measured; to be replaced by the ESE run).

Experiment (deferred to ESE)	Expected† (not measured)	Basis
DistilBERT full-train, WELFake test macro-F1 (O1)	≈0.98–0.99†; O1 (≥ 0.95) expected met†, not checked	20k-subset val 0.9863–0.9871 (Table VI); classical WELFake val–test gaps within ±0.003
DistilBERT, ISOT test (in-domain)	high†, no range	ISOT ⊂ WELFake; ∼80 % of ISOT test rows in WELFake train
DistilBERT, cross-dataset W\I ↔ ISOT	sharp drop, like the classical models† (direction only)	SVM test 0.6611 W\I→ISOT, 0.5480 ISOT→W\I
Fusion thresholds (dev-slice tuning)	$\tau = 0.6$, $N = 6$ months, $k = 5$†	current untuned fusion.yaml defaults
End-to-end 5-class, Live Claims Set (O6)	target†: macro-F1 ≥ 0.55, Unverifiable precision ≥ 0.7	project objective O6 (a target, not a prediction)

H. Published Reference Results (Other Work)

Table X places the deferred experiments next to results published by other authors. These are not our measurements: we copied them from the cited papers and model cards and did not reproduce any of them, and they do not fill the deferred cells of Tables VI and IX, which stay "deferred to ESE". They are not directly comparable to our protocol‡.

TABLE X
Published reference results of other work — not our measurement (copied from the cited sources; ‡see note).

Our deferred experiment	Published result (other work)	Setup of that work
DistilBERT, WELFake test	DistilBERT acc. 95.38 %, macro-F1 0.9538; BERT 0.9592; RoBERTa 0.9684 [21]	raw WELFake (72,134), stratified 80/10/10, test $n = 7{,}214$, max. len. 512
WELFake (reference model)	acc. 96.73 %, reported as +1.31 % over BERT [7]	WELFake paper's own model and split
DistilBERT, ISOT test (in-domain)	acc. (F1) DistilBERT 0.989 (0.989), BERT 0.990 (0.989), DeBERTa 0.997 (0.989) [22]	ISOT with only URLs/punctuation removed, 80 % train / 10 % test
DistilBERT, cross-dataset	ISOT→CC: DistilBERT 0.670 (0.655), BERT 0.670 (0.702); CC→ISOT: DistilBERT 0.775 (0.750) [22]	CC = Combined Corpus of Khan et al., not our W\I split
Stance model, in-domain NLI	nli-deberta-v3-small: SNLI test 91.65, MNLI mism. 87.55 acc. [42]; DeBERTa-v3-small MNLI-m/mm 88.3/87.7 [43]	model cards; NLI test sets, not FEVER (ours: 0.7267, Section VI)

‡The cited works train and test on raw WELFake/ISOT with their own random splits, without our artefact stripping (Reuters datelines, credits, captions, bylines, title tags) and without cross-corpus deduplication; WELFake contains ISOT, and artefacts alone separate raw ISOT at 0.9973 accuracy (Section III), so their scores can include leakage our protocol removes. Accuracy and F1 are as printed by each source.

VII. Discussion and Comparison with Existing Systems

VIII. Conclusion and Future Work

Acknowledgment

The authors thank their project guide and the Department of CSE (AI/ML), KIET Group of Institutions, for guidance and facilities.

References

[1] S. Vosoughi, D. Roy, and S. Aral, "The spread of true and false news online," Science, vol. 359, no. 6380, pp. 1146–1151, 2018.

[2] World Economic Forum, "The global risks report 2024," World Economic Forum, Geneva, Switzerland, Tech. Rep., Jan. 2024, 19th edition. https://www.weforum.org/publications/global-risks-report-2024/.

[3] M. S. Islam, T. Sarkar, S. H. Khan, A.-H. Mostofa Kamal, S. M. M. Hasan, A. Kabir, D. Yeasmin, M. A. Islam, K. I. Amin Chowdhury, K. S. Anwar, A. A. Chughtai, and H. Seale, "COVID-19–related infodemic and its impact on public health: A global social media analysis," The American Journal of Tropical Medicine and Hygiene, vol. 103, no. 4, pp. 1621–1629, 2020.

[4] D. Bathini, "How WhatsApp helped turn an Indian village into a lynch mob," BBC News, https://www.bbc.com/news/world-asia-india-44856910, Jul. 2018, 18 July 2018; accessed 28 Aug 2026.

[5] Al Jazeera, "In India, WhatsApp stirs up deadly rumours," https://www.aljazeera.com/features/2018/7/17/in-india-whatsapp-stirs-up-deadly-rumours, Jul. 2018, 17 July 2018; accessed 28 Aug 2026.

[6] M. S. Al-Zaman, "COVID-19-related social media fake news in India," Journalism and Media, vol. 2, no. 1, pp. 100–114, 2021.

[7] P. K. Verma, P. Agrawal, I. Amorim, and R. Prodan, "WELFake: Word embedding over linguistic features for fake news detection," IEEE Transactions on Computational Social Systems, vol. 8, no. 4, pp. 881–893, 2021.

[8] R. K. Kaliyar, A. Goswami, and P. Narang, "FakeBERT: Fake news detection in social media with a BERT-based deep learning approach," Multimedia Tools and Applications, vol. 80, pp. 11 765–11 788, 2021.

[9] H. Ahmed, I. Traore, and S. Saad, "Detection of online fake news using N-gram analysis and machine learning techniques," in Intelligent, Secure, and Dependable Systems in Distributed and Cloud Environments (ISDDC 2017), ser. Lecture Notes in Computer Science, vol. 10618. Springer, 2017, pp. 127–138.

[10] W. Y. Wang, ""Liar, Liar Pants on Fire": A new benchmark dataset for fake news detection," in Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers). Vancouver, Canada: Association for Computational Linguistics, 2017, pp. 422–426.

[11] H. Ahmed, I. Traore, and S. Saad, "Detecting opinion spams and fake news using text classification," Security and Privacy, vol. 1, no. 1, p. e9, 2018.

[12] K. Shu, D. Mahudeswaran, S. Wang, D. Lee, and H. Liu, "FakeNewsNet: A data repository with news content, social context, and spatiotemporal information for studying fake news on social media," Big Data, vol. 8, no. 3, pp. 171–188, 2020.

[13] J. Thorne, A. Vlachos, C. Christodoulopoulos, and A. Mittal, "FEVER: A large-scale dataset for fact extraction and VERification," in Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers). New Orleans, Louisiana: Association for Computational Linguistics, 2018, pp. 809–819.

[14] I. Augenstein, C. Lioma, D. Wang, L. Chaves Lima, C. Hansen, C. Hansen, and J. G. Simonsen, "MultiFC: A real-world multi-domain dataset for evidence-based fact checking of claims," in Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP). Hong Kong, China: Association for Computational Linguistics, 2019, pp. 4685–4697.

[15] F. Arslan, N. Hassan, C. Li, and M. Tremayne, "A benchmark dataset of check-worthy factual claims," in Proceedings of the Fourteenth International AAAI Conference on Web and Social Media (ICWSM), vol. 14, 2020, pp. 821–829.

[16] X. Zhou and R. Zafarani, "A survey of fake news: Fundamental theories, detection methods, and opportunities," ACM Computing Surveys, vol. 53, no. 5, pp. 109:1–109:40, 2020.

[17] Z. Guo, M. Schlichtkrull, and A. Vlachos, "A survey on automated fact-checking," Transactions of the Association for Computational Linguistics, vol. 10, pp. 178–206, 2022.

[18] V. Pérez-Rosas, B. Kleinberg, A. Lefevre, and R. Mihalcea, "Automatic detection of fake news," in Proceedings of the 27th International Conference on Computational Linguistics (COLING). Santa Fe, New Mexico, USA: Association for Computational Linguistics, 2018, pp. 3391–3401.

[19] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers). Minneapolis, Minnesota: Association for Computational Linguistics, 2019, pp. 4171–4186.

[20] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," arXiv preprint arXiv:1910.01108, 2019, presented at the 5th Workshop on Energy Efficient Machine Learning and Cognitive Computing (NeurIPS 2019).

[21] M. T. S. Abdo and H. M. Roki, "Transformer vs. classical ML for fake news detection: A SHAP-explainable comparative study on WELFake," International Journal of Computer Applications, vol. 187, no. 119, Jun. 2026. [Online]. Available: https://ijcaonline.org/archives/volume187/number119/abdo-ijca-2026-464629276ffa.pdf

[22] C. Blackledge and A. Atapour-Abarghouei, "Transforming fake news: Robust generalisable news classification using transformers," 2021. [Online]. Available: https://arxiv.org/abs/2109.09796

[23] N. Hassan, F. Arslan, C. Li, and M. Tremayne, "Toward automated fact-checking: Detecting check-worthy factual claims by ClaimBuster," in Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. Halifax, NS, Canada: ACM, 2017, pp. 1803–1812.

[24] Y. Nie, H. Chen, and M. Bansal, "Combining fact extraction and verification with neural semantic matching networks," in Proceedings of the AAAI Conference on Artificial Intelligence, vol. 33, no. 01, 2019, pp. 6859–6866.

[25] Z. Liu, C. Xiong, M. Sun, and Z. Liu, "Fine-grained fact verification with kernel graph attention network," in Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, 2020, pp. 7342–7351.

[26] K. Popat, S. Mukherjee, A. Yates, and G. Weikum, "DeClarE: Debunking fake news and false claims using evidence-aware deep learning," in Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing. Brussels, Belgium: Association for Computational Linguistics, 2018, pp. 22–32.

[27] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP). Hong Kong, China: Association for Computational Linguistics, 2019, pp. 3982–3992.

[28] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," in Advances in Neural Information Processing Systems, vol. 33, 2020, pp. 5776–5788.

[29] P. He, J. Gao, and W. Chen, "DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing," in The Eleventh International Conference on Learning Representations (ICLR), 2023.

[30] A. Williams, N. Nangia, and S. Bowman, "A broad-coverage challenge corpus for sentence understanding through inference," in Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers). New Orleans, Louisiana: Association for Computational Linguistics, 2018, pp. 1112–1122.

[31] A. Hanselowski, A. PVS, B. Schiller, F. Caspelherr, D. Chaudhuri, C. M. Meyer, and I. Gurevych, "A retrospective analysis of the fake news challenge stance-detection task," in Proceedings of the 27th International Conference on Computational Linguistics (COLING). Santa Fe, New Mexico, USA: Association for Computational Linguistics, 2018, pp. 1859–1874.

[32] M. T. Ribeiro, S. Singh, and C. Guestrin, ""Why should I trust you?": Explaining the predictions of any classifier," in Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. San Francisco, CA, USA: ACM, 2016, pp. 1135–1144.

[33] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in Neural Information Processing Systems, vol. 30, 2017, pp. 4765–4774.

[34] K. Shu, L. Cui, S. Wang, D. Lee, and H. Liu, "dEFEND: Explainable fake news detection," in Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. Anchorage, AK, USA: ACM, 2019, pp. 395–405.

[35] A. Zubiaga, A. Aker, K. Bontcheva, M. Liakata, and R. Procter, "Detection and resolution of rumours in social media: A survey," ACM Computing Surveys, vol. 51, no. 2, pp. 32:1–32:36, 2018.

[36] A. Barbaresi, "Trafilatura: A web scraping library and command-line tool for text discovery and extraction," in Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing: System Demonstrations. Association for Computational Linguistics, 2021, pp. 122–131.

[37] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," IEEE Transactions on Big Data, vol. 7, no. 3, pp. 535–547, 2021.

[38] S. R. Bowman, G. Angeli, C. Potts, and C. D. Manning, "A large annotated corpus for learning natural language inference," in Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing. Lisbon, Portugal: Association for Computational Linguistics, 2015, pp. 632–642.

[39] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[40] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. Le Scao, S. Gugger, M. Drame, Q. Lhoest, and A. M. Rush, "Transformers: State-of-the-art natural language processing," in Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations. Association for Computational Linguistics, 2020, pp. 38–45.

[41] C. Malon, "Team Papelo: Transformer networks at FEVER," in Proceedings of the First Workshop on Fact Extraction and VERification (FEVER). Brussels, Belgium: Association for Computational Linguistics, 2018, pp. 109–113.

[42] Sentence-Transformers, "cross-encoder/nli-deberta-v3-small (model card)," Hugging Face Hub, accessed 2026-10-26. [Online]. Available: https://huggingface.co/cross-encoder/nli-deberta-v3-small

[43] Microsoft, "microsoft/deberta-v3-small (model card)," Hugging Face Hub, accessed 2026-10-26. [Online]. Available: https://huggingface.co/microsoft/deberta-v3-small